

2016.0 RANGE ROVER (LG), 310-01

FUEL TANK AND LINES - TDV6 3.0L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the fuel tank and lines system and operation, refer to the relevant Description and Operation section of the workshop manual.

REFER to: [Fuel Tank and Lines](#) (310-01A Fuel Tank and Lines - TDV6 3.0L Diesel, Description and Operation).

INSPECTION AND VERIFICATION

WARNINGS:

- Do **NOT** carry out any work on the fuel system with the engine running. The fuel pressure within the system can be as high as 2000 bar (29,008 lb/in²). Failure to follow this instruction may result in personal injury.
- Eye protection must be worn at all times when working on or near any fuel related components. Failure to follow this instruction may result in personal injury.
- This procedure involves fuel handling. Be prepared for fuel spillage at all times and always observe fuel handling precautions. Failure to follow this instruction may result in personal injury.
- After carrying out repairs, the fuel system must be checked visually for leaks. This should be done after the engine has been run, but with the engine switched **OFF**. Failure to follow this instruction may result in personal injury.
- If taken internally, **DO NOT** induce vomiting. Seek immediate medical attention. Failure to follow this instruction may result in personal injury.
- If fuel contacts the eyes, flush the eyes with cold water or eyewash solution and seek medical attention. Failure to follow this instruction may result in personal injury.
- Wash hands thoroughly after handling, as prolonged contact may cause irritation. Should irritation develop, seek medical attention. Failure to follow this instruction may result in personal injury.

CAUTIONS:

- Before disconnecting any part of the system, it is imperative that all dust, dirt and debris is removed from around components to prevent ingress of foreign matter into the fuel system. Failure to

follow this instruction may result in damage to the vehicle.

- The fuel pipes between the injectors and the rail must be discarded after each use, and new pipes installed. Failure to follow this instruction may result in damage to the vehicle.
- It is essential that absolute cleanliness is observed when working with these components. Always install blanking plugs to any open orifices or lines. Failure to follow this instruction may result in damage to the vehicle.
- Make sure that the workshop area in which the vehicle is being worked on is as clean and dust-free as possible. Areas in which work on clutches, brakes or where welding or machining are carried out are not suitable in view of the risk of contamination to the fuel system. Failure to follow this instruction may result in damage to the vehicle.
- Make sure that any protective clothing worn is clean and made from lint-free non-flocking material. Failure to follow this instruction may result in damage to the vehicle.
- Make sure that any protective gloves worn are new and are of the non-powdered latex type. Failure to follow this instruction may result in damage to the vehicle.
- Make sure that clean, non-plated tools are used. Clean tools using a new brush that will not lose its bristles and fresh cleaning fluid prior to starting work on the vehicle. Failure to follow this instruction may result in damage to the vehicle.
- Use a steel-topped work bench and cover it with clean, lint-free, non-flocking material. Failure to follow this instruction may result in damage to the vehicle.
- Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTES:

- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- When measuring fuel sender resistance values with a multimeter, it is critical to use the correct multimeter setting. The multimeter should **NOT** be on the 'Auto' setting and **MUST** be set to 'Manual'. This will help prevent incorrect diagnosis and unnecessary replacement of fuel senders. If the multimeter range is set at 'Auto' then, during a sweep of the sender from 50 Ohms to 998 Ohms, the multimeter has to change its measurement range. For approximately 1 second, during the range switch over point, the multimeter display indicates an open circuit. This can lead to a mis-diagnosis of a fuel sender fault.

1. Verify the customer concern.

2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Fuel level ▪ Contaminated fuel ▪ Fuel supply line(s) ▪ Fuel return line(s) ▪ High-pressure fuel supply line(s) ▪ Fuel tank filler pipe ▪ Fuel leak(s) ▪ Fuel tank ▪ Fuel filler cap ▪ Fuel filter ▪ Push connect fittings ▪ Fuel rail	▪ Battery charge and condition ▪ Fuse(s) ▪ Fuel pump module relay ▪ Fuel pump module ▪ Electrical connector(s) ▪ Damaged or corroded wiring harness ▪ Fuel volume control valve ▪ Fuel pressure control valve ▪ Engine control module

- Fuel injection pump
- Exhaust gas recirculation system

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Engine cranks, but does not start	▪ Low/contaminated fuel ▪ Air leakage ▪ Low-pressure fuel system fault ▪ Fuel pump module (lift pump) fault ▪ Blocked fuel filter ▪ Fuel volume regulator blocked/contaminated ▪ Fuel pressure control valve blocked/contaminated ▪ Fuel pump fault ▪ Crankshaft position sensor	Check the fuel level and condition. Check the low pressure fuel pump operation. Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the intake air system for leaks. Check the lift pump operation, check the low-pressure fuel system for leaks/damage. Check the fuel filter, check for DTCs indicating a fuel volume or pressure control valve fault. Check the fuel pump. Check the crankshaft position sensor circuits. Refer to the electrical guides
Difficult to start	▪ Glow plug system fault (very cold conditions) ▪ Low/contaminated fuel ▪ Air leakage ▪ Fuel pump module	Check the glow plug circuits. Refer to the electrical guides. Check the fuel level/condition. Check the low pressure fuel pump operation. Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no

	(lift pump) fault ▪ Low-pressure fuel system fault ▪ Blocked fuel filter ▪ Fuel volume control valve blocked/contaminated ▪ Fuel pressure control valve blocked/contaminated ▪ Exhaust gas recirculation valve(s) fault	separation of the fuel indicating water or other liquid in the fuel. Check the intake air system for leaks. Check the lift pump operation, check the low-pressure fuel system for leaks/damage. Check the fuel filter, check for DTCs indicating a fuel volume or pressure control valve fault. Check the exhaust gas recirculation system
Rough idle	▪ Intake air system fault ▪ Low/contaminated fuel ▪ Low-pressure fuel system fault ▪ Blocked fuel filter ▪ Fuel volume control valve blocked/contaminated ▪ Fuel pressure control valve blocked/contaminated ▪ Exhaust gas recirculation valve(s) fault	Check the intake air system for leaks. Check the low pressure fuel pump operation. Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump. Check the fuel level/condition. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the low-pressure fuel system for leaks/damage. Check the fuel filter, check for DTCs indicating a fuel volume or pressure control valve fault. Check the exhaust gas recirculation system
Lack of power when accelerating	▪ Intake air system fault ▪ Restricted exhaust system ▪ Low fuel pressure ▪ Exhaust gas recirculation valve(s) fault ▪ Turbocharger actuator fault	Check the intake air system for leakage or restriction. Check for a blockage/restriction in the exhaust system, install new components as necessary. Check the low pressure fuel pump operation. Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump. Check for DTCs indicating a fuel pressure fault. Check the exhaust gas recirculation system. Check turbocharger actuator
Engine stops/stalls	▪ Air leakage ▪ Low/contaminated	Check the intake air system for leaks. Check the fuel level/condition. Using the

	fuel ▪ Low-pressure fuel system fault ▪ High pressure fuel leak ▪ Fuel volume control valve blocked/contaminated ▪ Fuel pressure control valve blocked/contaminated ▪ Exhaust gas recirculation valve(s) fault	manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the fuel system for leaks/damage: Check for DTCs indicating a fuel volume or pressure control valve fault. Check the exhaust gas recirculation system
Engine judders	▪ Low/contaminated fuel ▪ Air ingress ▪ Low-pressure fuel system fault ▪ Fuel metering valve blocked/contaminated ▪ Fuel volume control valve blocked/contaminated ▪ Fuel pressure control valve blocked/contaminated ▪ High pressure fuel leak ▪ Fuel pump fault	Check the fuel level/condition. Check the low pressure fuel pump operation. Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the intake air system for leaks. Check the low-pressure fuel system for leaks/damage. Check the high pressure fuel system for leaks, check for DTCs indicating a fuel volume or pressure control valve fault. Check the fuel pump
Excessive fuel consumption	▪ Low-pressure fuel system fault ▪ Fuel volume control valve blocked/contaminated ▪ Fuel pressure control valve blocked/contaminated ▪ Fuel temperature sensor leak ▪ High pressure fuel	Check the low-pressure fuel system for leaks/damage. Check the low pressure fuel pump operation. Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump. Check for DTCs indicating a fuel volume or pressure control valve fault. Check the fuel temperature sensor, fuel pump, etc for leaks. Check for injector DTCs. Check the exhaust gas recirculation system

	leak	
	▪ Injector(s) fault ▪ Exhaust gas recirculation valve(s) fault	

MIS-FUELLING ISSUES / ACCIDENTALLY FILLING THE FUEL TANK WITH GASOLINE INSTEAD OF DIESEL

NOTES:

- The customer **MUST** be informed prior to vehicle hand-over of the consequences of adding gasoline to the fuel tank instead of diesel.
- The cost of this repair work is not Jaguar Land Rover's responsibility. If the customer does not have all of the appropriate work carried out, warranty may be refused on all affected components.
- If the customer refuses the advised repair or wants a lower level repair than advised, the vehicle must be added to the Restricted Warranty Register.
- Vehicle mis-fuelling will not result in a formal warranty restriction if the approved repair has been completed.
- The symptom charts below describe the four possible scenarios of a customer incorrectly fuelling a diesel engine vehicle with gasoline. When customers report a mis-fuel, allow them to fully explain exactly what took place before advising the necessary repair.
- The procedures listed below are for guidance only. The relevant sections of TOPlx must be referred to for all detailed procedures and replacement of parts.
- Purge operation of the low pressure fuel pumps on the following vehicle lines is activated when the drivers door is opened: Jaguar XE, Jaguar XF, Jaguar F-Pace. For all other vehicle lines, the purge operation of the low pressure fuel pumps is activated when the ignition is switched on.

Mis-fuelling Procedure 1

SYMPTOM	POSSIBLE CAUSES	ACTION
Mis-fuelling issue. The diesel fuel tank has been filled with gasoline	The vehicle has been incorrectly fuelled and the ignition switch has NOT been switched to ON	▪ 1. Referring to the relevant section in the workshop manual, drain the fuel tank

		▪ 2. Remove the fuel tank assembly ▪ 3. Remove the primary fuel pump/sender module and in-tank transfer pipe. Invert the pump/sender module to drain excess fuel ▪ 4. Remove all traces of residual fuel within the fuel tank using clean absorbent cloth/paper ▪ 5. Flush the pump/sender module and in tank transfer pipe with clean diesel ▪ 6. Reassemble the pump/sender module and in tank transfer pipe to the fuel tank and install to the vehicle ▪ 7. Fully fill the fuel tank with clean diesel ▪ 8. Road test the vehicle for at least 10 miles/16Km ▪ 9. Return the vehicle to the customer
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Mis-fuelling Procedure 2

SYMPTOM	POSSIBLE CAUSES	ACTION
Mis-fuelling issue. The diesel fuel tank has been filled with gasoline	The vehicle has been incorrectly fuelled and the ignition switch has been switched to the ON position	▪ 1. Referring to the relevant section in the workshop manual, drain the fuel tank ▪ 2. Remove the fuel pump module and fuel level sensor ▪ 3. Remove all traces of residual fuel within the fuel tank using clean absorbent cloth/paper ▪ 4. Invert the fuel pump module and fuel level sensor to remove all traces of residual fuel ▪ 5. Remove the fuel filter and discard ▪ 6. Disconnect the fuel feed to the high-pressure fuel pump ▪ 7. Flush all the fuel lines with

		clean diesel ▪ 8. Reassemble the fuel pump module and fuel level sensor to the fuel tank ▪ 9. Install a new fuel filter ▪ 10. Fully fill the fuel tank with clean diesel ▪ 11. Road test the vehicle for at least 10 miles/16Km, checking for normal engine operation and no stored DTC'S ▪ 12. Return the vehicle to the customer
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Mis-fuelling Procedure 3

SYMPTOM	POSSIBLE CAUSES	ACTION
Mis-fuelling issue. The diesel fuel tank has been filled with gasoline	The vehicle has run for less than two minutes	▪ 1. Referring to the relevant section in the workshop manual, drain the fuel tank ▪ 2. The residual fuel from all removed parts needs to be checked for metal particles using a magnet ▪ 3. Remove the fuel pump module and fuel level sensor ▪ 4. Remove all traces of residual fuel within the fuel tank using clean absorbent cloth/paper ▪ 5. Invert the fuel pump module and fuel level sensor to remove all traces of residual fuel ▪ 6. Remove the fuel filter and discard ▪ 7. Disconnect the fuel feed to the high-pressure fuel pump ▪ 8. Flush all the fuel lines with clean diesel ▪ 9. If no metal particles are detected, go to step 11 ▪ 10. If metal particles are present, go to procedure 4 and carry out step 3 to step 8

		▪ 11. Reassemble the fuel pump module and fuel level sensor to the fuel tank ▪ 12. Install a new fuel filter ▪ 13. Fully fill the fuel tank with clean diesel ▪ 14. Road test the vehicle for at least 10 miles/16Km, checking for normal engine operation and no DTC'S ▪ 15. Return the vehicle to the customer
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Mis-fuelling Procedure 4

SYMPTOM	POSSIBLE CAUSES	ACTION
Mis-fuelling issue. The diesel fuel tank has been filled with gasoline	The vehicle has been driven until it stops or over two minutes. The entire fuel system will need reworking	▪ 1. Referring to the relevant section in the workshop manual, drain the fuel tank ▪ 2. Clean and flush all the low pressure fuel pipes ▪ 3. Replace the fuel filter ▪ 4. Replace the high pressure fuel pump ▪ 5. Replace all the high pressure fuel pipes ▪ 6. Replace the diverter pipe and both fuel rails ▪ 7. Replace the fuel cooler ▪ 8. Replace all the fuel injectors ▪ 9. Remove the fuel pump module and fuel level sensor ▪ 10. Remove all the traces of residual fuel within the fuel tank using clean absorbent cloth/paper ▪ 11. Invert the fuel pump module and fuel level sensor to remove all the traces of residual fuel ▪ 12. Reassemble the fuel pump module and fuel level sensor to the fuel tank ▪ 13. Fully fill the fuel tank with clean diesel

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| | | <ul style="list-style-type: none">▪ 14. Road test the vehicle for at least 10 miles/16Km, checking for normal engine operation and that no stored diagnostic trouble codes are logged▪ 15. Return the vehicle to the customer |
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D T C I N D E X

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5, DTC: Engine Control Module \(ECM\)](#) (100-00 General Information, Description and Operation).